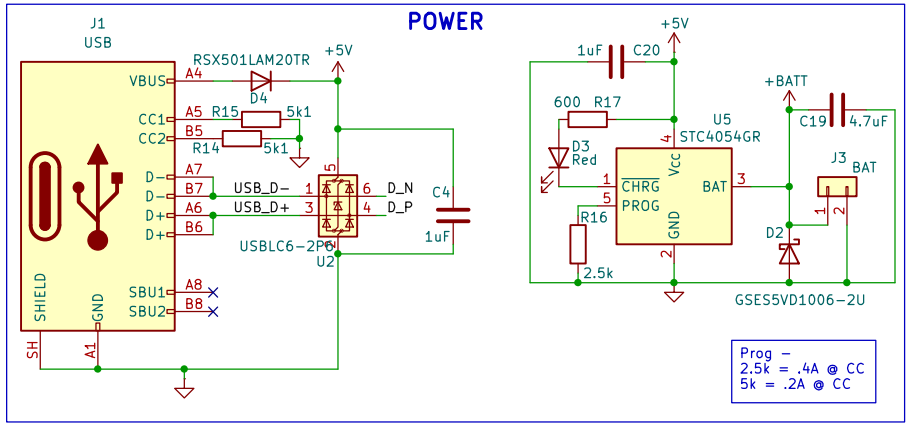
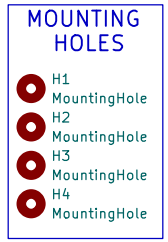
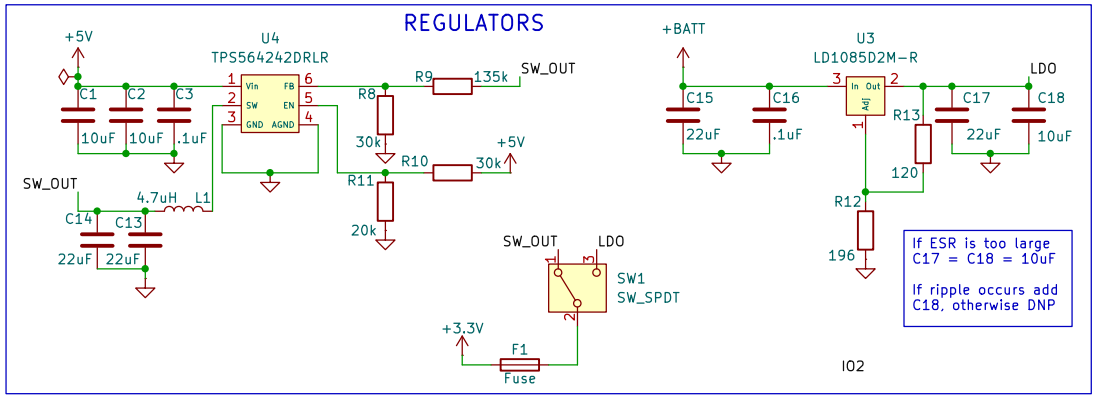
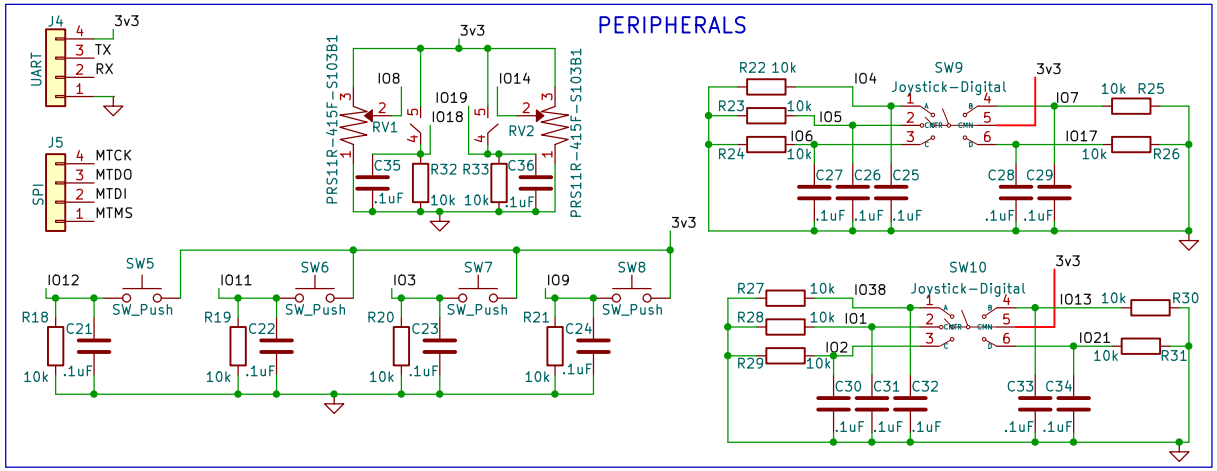
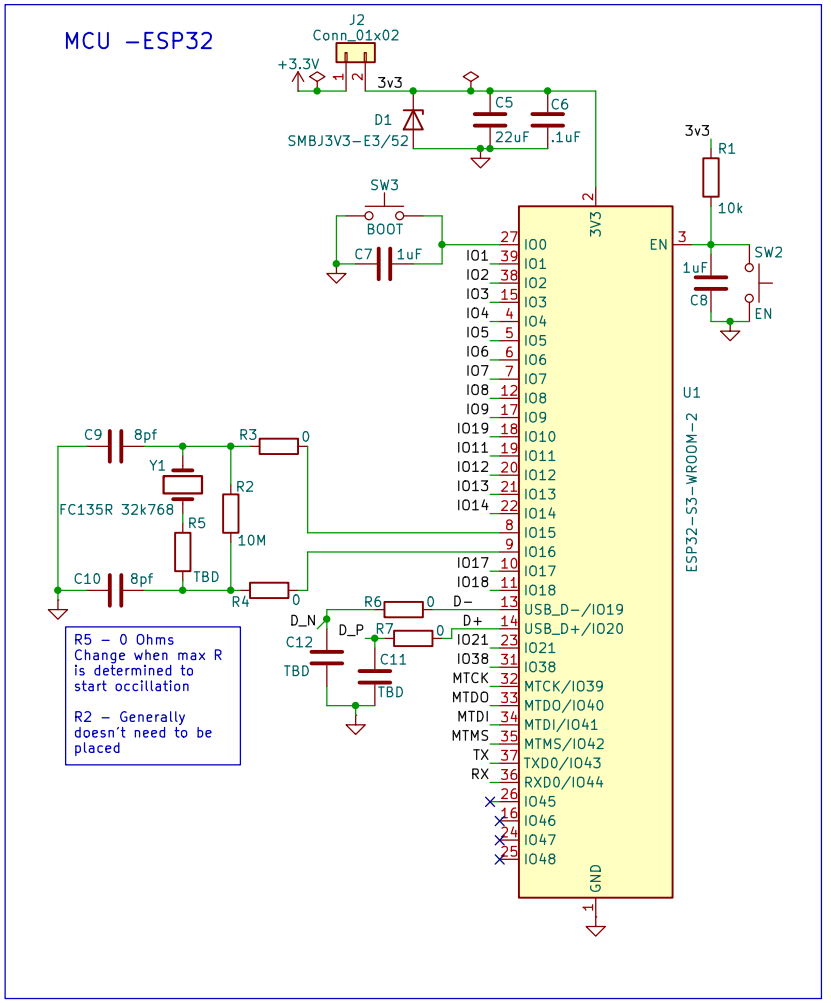

POWER
J1
USB
VBUS
CC1
CC2
D-
D-
D+
D+
SHIELD
GND
SBU1
SBU2
SH
A1
RSX501LAM20TR
D4
+5V
R15 5k1
R14 5k1
USB_D-
USB_D+
USBLC6-2P6
U2
D_N
D_P
C4
1uF
C20
1uF
600 R17
D3
Red
U5
STC4054GR
CHRG
PROG
VCC
BAT
GND
R16
2.5k
+BATT
C19 4.7uF
J3
BAT
D2
GSES5VD1006-2U
Prog -
2.5k = .4A @ CC
5k = .2A @ CC


MOUNTING HOLES
H1
MountingHole
H2
MountingHole
H3
MountingHole
H4
MountingHole


REGULATORS
+5V
U4
TPS564242DRLR
C1
C2
C3
10uF
10uF
.1uF
SW_OUT
4.7uH
L1
C14
C13
22uF
22uF
R9 135k
R8
30k
R10 30k
R11
20k
+BATT
U3
LD1085D2M-R
C15
22uF
C16
.1uF
C17
22uF
C18
10uF
LDO
R13
120
R12
196
If ESR is too large
C17 = C18 = 10uF
If ripple occurs add
C18, otherwise DNP
SW1
SW_SPDT
+3.3V
F1
Fuse
IO2


PERIPHERALS
J4
UART
TX
RX
3v3
J5
SPI
MTCK
MTDO
MTDI
MTMS
PRS11R-415F-S103B1
RV1
RV2
IO8
IO19
IO18
IO14
C35
.1uF
R32 R33
10k 10k
C36
.1uF
R22 10k
R23 10k
R24 10k
IO4
IO5
IO6
SW9
Joystick-Digital
IO7
10k R25
IO17
10k R26
C27 C26 C25
.1uF .1uF .1uF
C28 C29
.1uF .1uF
SW5
SW6
SW7
SW8
IO12
IO11
IO3
IO9
SW_Push
R18 10k
R19 10k
R20 10k
R21 10k
C21 .1uF
C22 .1uF
C23 .1uF
C24 .1uF
R27 10k
R28 10k
R29 10k
IO38
IO1
IO2
SW10
Joystick-Digital
IO13 10k R30
IO21
10k R31
C30 C31 C32
.1uF .1uF .1uF
C33 C34
.1uF .1uF


MCU -ESP32
J2
Conn_01x02
+3.3V
3v3
D1
SMBJ3V3-E3/52
C5
22uF
C6
.1uF
R1
10k
SW3
BOOT
C7 1uF
SW2
C8 1uF
EN
U1
ESP32-S3-WROOM-2
C9 8pf
Y1
FC135R 32k768
C10 8pf
R5
TBD
R3 0
R2
10M
R4 0
R6 0
R7 0
D_N
D_P
C12
TBD
C11
TBD
R5 - 0 Ohms
Change when max R
is determined to
start occillation
R2 - Generally
doesn't need to be
placed
IO0
IO1
IO2
IO3
IO4
IO5
IO6
IO7
IO8
IO9
IO19
IO11
IO12
IO13
IO14
IO15
IO16
IO17
IO18
D-
D+
IO21
IO38
MTCK
MTDO
MTDI
MTMS
TX
RX
IO45
IO46
IO47
IO48
USB_D-/IO19
USB_D+/IO20
MTCK/IO39
MTDO/IO40
MTDI/IO41
MTMS/IO42
TXD0/IO43
RXD0/IO44
GND